A table-free linear 2 + 1 decomposition on the frozen m = 3 cells

Study `research/groebner_free_matched_20261113`, run 2026-11-13T15:41Z, protocol frozen in commit `c90b1d9d54`, benchmark record `20261113T1539Z-ic-m3-linear21-vs-floor`. Stage diagnostic; every number is `[pcc:unverified]` (one session, shared 4-vCPU Xeon container, isolation L0, one core, no validator replay).

Result

On the same cells and targets as `ic_m3_algebraic_vs_mitm_20261014` (seed 20261014), plus five larger cells, the arm `lin21-ht` decides three-summand decomposability with no Gröbner basis and no pair table. Measured:

- **Exactness.** The floor's verdict on 16,384 of 16,384 pool targets (eight cells with a pair table); 264 of 264 child solves agree and every witness adds up in the group.
- **Counted work.** Slope of $\log_2$ work per relation on n: trials 0.3352 [0.3311, 0.3397], floor additions 0.3346 [0.3313, 0.3382], difference 0.0006 [−0.0004, 0.0016].
- **Wall time.** 2.89 to 10.87 times the floor's online cost per relation; under the frozen 2026-10-14 rule the wall-time slope difference is [0.040, 0.057] ("grows faster"), which is one trial rising from about 250 ns to 500 ns over n = 13 to 41.
- **Memory.** 15.7 MB against 2,558 MB at n = 41. It grows with the base: 519 MB at n = 59 (the prediction "under 100 MB on every cell" failed at n = 53 and 59).
- **Against F4 on the S_3 chain**, rerun in this session: 375 times cheaper per relation at n = 13, 8,500 times at n = 31.
- **Range.** Runs to n = 59, where the pair table would hold $5.5 \cdot 10^{11}$ entries.

It does not grow more slowly than the floor: it tries every base point as third summand.

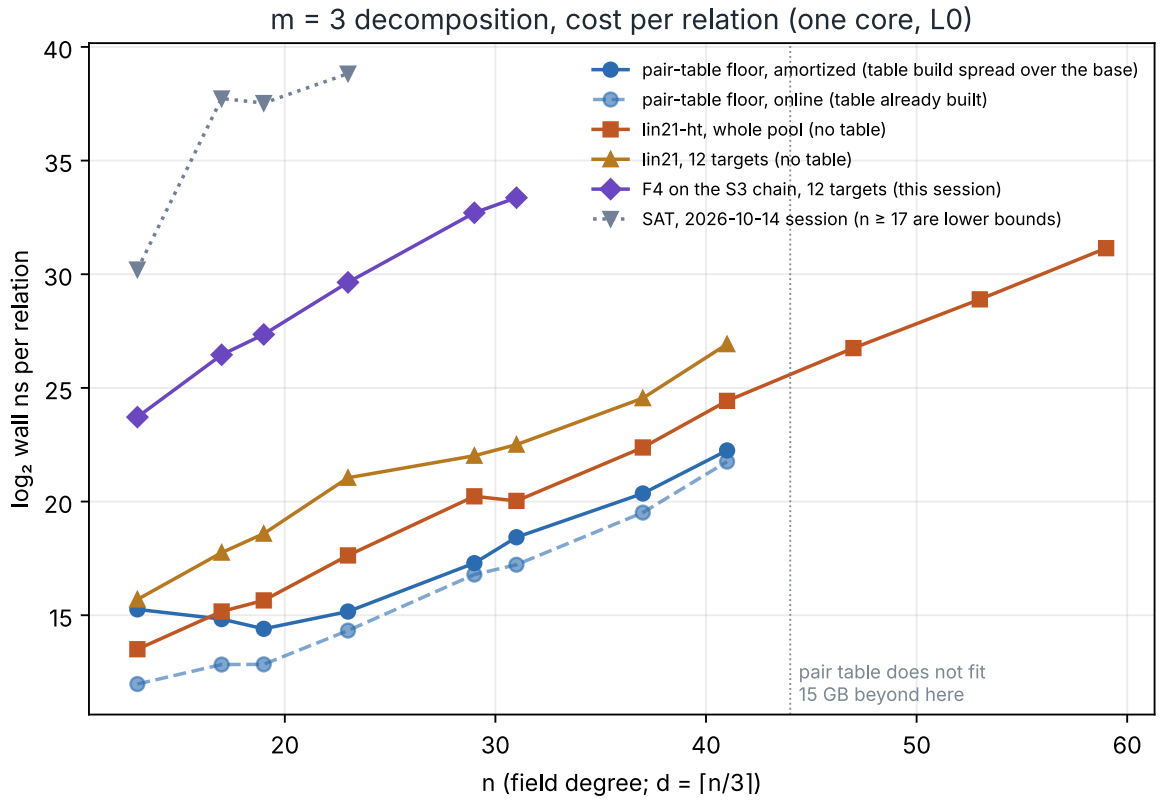


Figure 1: Cost per relation, $\log_2$ wall ns. Floor and `lin21-ht` over whole pools (2,048 targets; 1,024 at n = 53; 512 at n = 59); `lin21` and F4 on 12 targets per cell. SAT points are from the 2026-10-14 session. Run 20261113T1541Z.

Whole pool

n	d	|F|	p	lin ÷ floor online [95%]	amortized [95%]	ns/trial	floor MB	lin MB
13	5	33	0.494	2.89 [2.66, 3.13]	0.36 [0.34, 0.38]	280	6.9	8.3
17	6	63	0.274	5.04 [4.78, 5.29]	1.32 [1.23, 1.41]	202	6.9	8.2
19	7	117	0.404	7.03 [6.23, 8.10]	2.46 [2.21, 2.82]	259	7.4	8.4
23	8	237	0.247	9.88 [9.57, 10.21]	5.56 [5.28, 5.88]	260	8.2	8.4

29	10	1,017	0.279	10.87 [10.60, 11.13]	7.68 [7.37, 7.98]	430	19.5	8.9
31	11	2,101	0.479	7.00 [6.79, 7.22]	3.03 [2.87, 3.20]	394	70.8	9.1
37	13	8,191	0.488	7.28 [7.10, 7.46]	4.07 [3.89, 4.25]	528	722.8	12.0
41	14	16,223	0.277	6.37 [6.27, 6.48]	4.53 [4.38, 4.70]	489	2,558.1	15.7
47	16	65,363	0.284	no table		627		36.3
53	18	262,523	0.287	no table		697		132.9
59	20	1,049,853	0.287	no table		839		518.7

Measured values from `SUMMARY.json` (`ic_algebraic_floor linsummary`); intervals are a paired bootstrap over targets, 2,000 replicates. Zero disagreements on every floor cell.

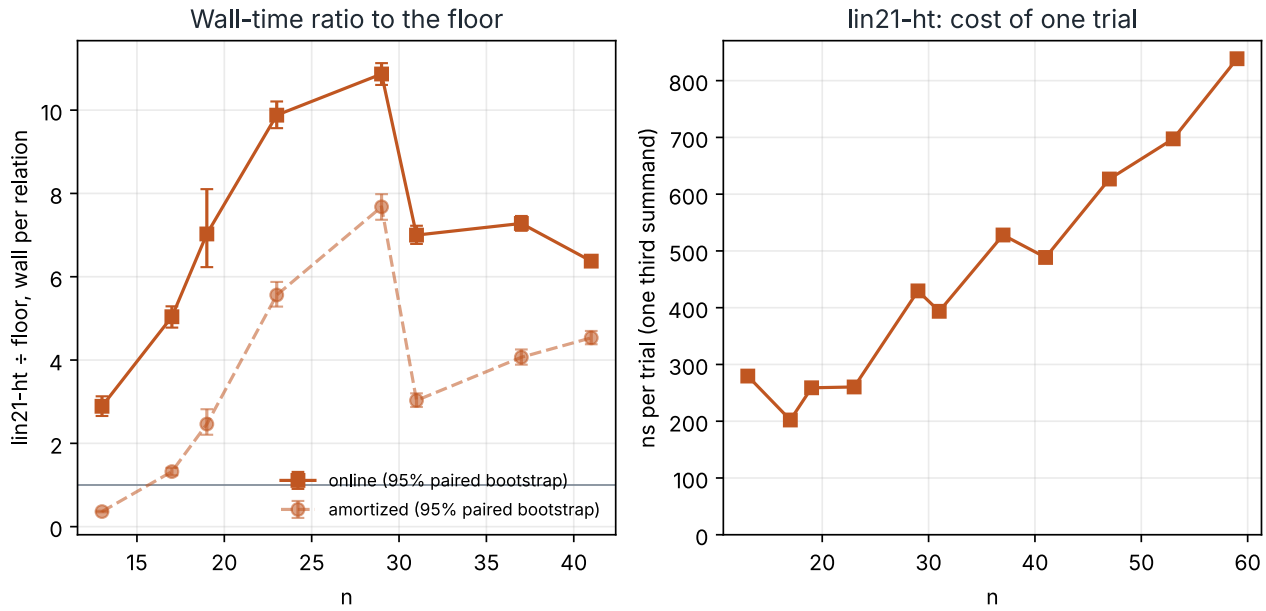


Figure 2: Left: wall-time ratio to the floor per relation. Right: cost of one trial of `lin21-ht`.

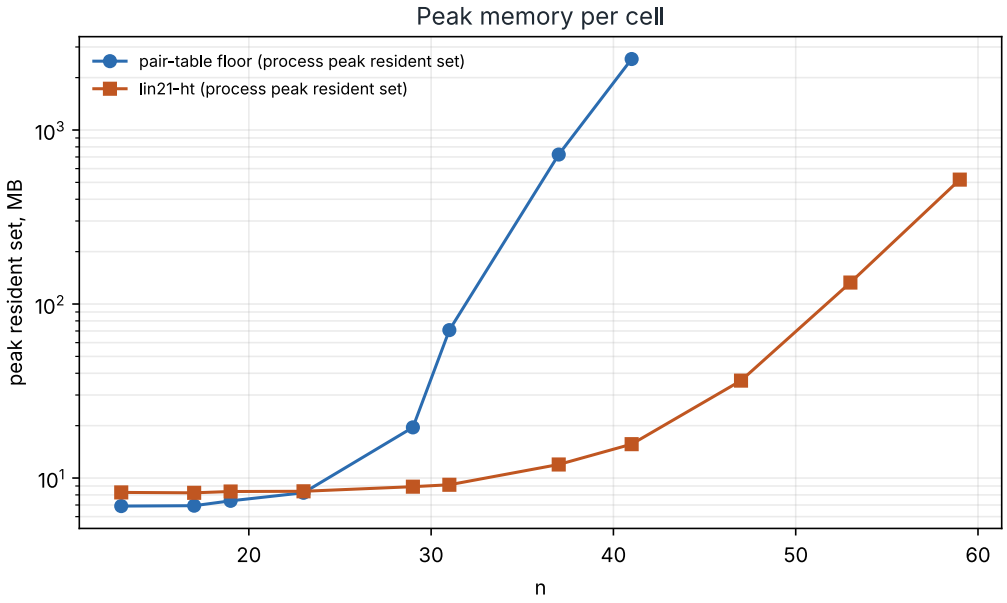


Figure 3: Peak resident set per cell (`VmHWM`), one cell per process.

The arm

With $s = x_1 + x_2$, $p = x_1x_2$ and $u = x(R - P_3)$, $S_3 = p^2 + u{\cdot}p + u^2s^2 + b$ is F_2-affine in (s, p); this is prior art in the repository (`ic_m2_linear_ladder_20261024`). `lin21-ht` eliminates p by the half-trace, $p = u{\cdot}(s + H(s) + H(b/u^2) + \varepsilon)$, leaving $n - 2d + 2$ equations in $d + 1$ bits, and splits by $x_1 = s{\cdot}H(p/s^2)$.

What stays open

To tie rho, $m \geq 4$ and one scan of the base per relation at $m = 4$; the floor and the $2 + 1 + 1$ form of this arm cost $|F|^2$. The missing piece is a three-summand decision with no scan. In the symmetric functions of the three abscissae, S_4 is F_2-affine except for $e_3 \cdot (u \cdot e_3^2 + u^3 e_2^2 + u^2 e_1^2 e_3 + b \cdot u \cdot e_1^2)$; `s4check` confirmed the identity on 2,095 witnesses with 0 of 2,095 control zeros. Plain linearisation of those four monomials reaches $d < n/31$ against $d \approx n/3$ needed.

Limits

One session at L0; the no-table curves have cofactors 1,318, 428 and 22,894; “no table” at $n \geq 47$ is by size, not a failed build; prior art was searched in the checkouts because the knowledge base is not installed here; no validator replay, so no benchmarks or leaderboard row. Raw records: `runs/`, `runs.log`; binary SHA-256 in `runs/binary.sha256`.